



Hands-On Workshop

Setting Up UDG

Prerequisites

Ensure the following are ready:

- Tyk Dashboard access
- Node.js v13+
- Docker Desktop or a local dev environment
- Git

Clone and Run Example REST APIs

We'll first set up the example services locally.

```
git clone https://github.com/example-org/example-rest-api-for-udg.git
cd example-rest-api-for-udg
npm install
npm run build
npm start
```

You should see:

Users Service Running on <http://localhost:4000>

Review service running on <http://localhost:4001>

Create a UDG API in Tyk Dashboard

- Go to Tyk Dashboard → APIs
- Click "Add New API"
- Select UDG (Universal Data Graph) as the API type
- Name it something like user-reviews-demo
- Set Authentication to "Keyless" for testing

Define Your GraphQL Schema

In the Schema tab, replace the default schema with:

```
type Mutation {  
  default: String  
}  
  
type Query {  
  user(id: String): User  
}  
  
type Review {  
  id: String  
  text: String  
  userId: String  
  user: User  
}  
  
type User {  
  id: String  
  username: String  
  reviews: [Review]  
}
```

Connect the user Query to the Users REST API

1. In the **Schema** tab, select the field `user` under `Query`
2. Choose Data Source Type → `REST`
3. Choose Use **External Data Source**
4. Fill in:
 - a. Datasource Name: `getUserById`
 - b. HTTP Method: `GET`
 - c. URL: `http://localhost:4000/users/{{.args.id}}`
(This uses GraphQL args as a dynamic path param)
1. (Optional) Add headers if needed
2. Save & Update API

Test the API in GraphQL Playground

Navigate to the Playground tab in Tyk Dashboard and run:

```
query getUser {  
  user(id: "1") {  
    username  
    id  
    reviews {  
      text  
    }  
  }  
}
```

Expected Response:

```
{  
  "data": {  
    "user": {  
      "username": "User 1",  
      "id": "1",  
      "reviews": null  
    }  
  }  
}
```

At this point, reviews is null because it's not yet connected.

You'll need to attach another REST data source for the reviews field next.

Challenge

- Try to resolve reviews field on type `Users`
- Try to resolve users field on type `Reviews`